

VINCENZO POMPONI

Robot Learning Researcher | Diffusion Policies · Imitation Learning · Vision for Human–Robot Collaboration

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Italian (EU) citizen · Swiss permit [B] · Available from September 2027

SUMMARY

PhD researcher at USI and the SUPSI ARM Lab developing generative robot policies that combine diffusion policies and Dynamic Movement Primitives for robust, generalizable manipulation. In parallel, builds vision-based activity recognition for human–robot collaboration in the assembly of heavy industrial components. Experienced in taking methods from research to real robots within EU-funded projects, with an engineering background from Politecnico di Milano.

EXPERIENCE

Scientific Collaborator & PhD Candidate

2022 – Present

SUPSI – Automation, Robotics and Machines (ARM) Laboratory / USI, Lugano, Switzerland

Generative robot policies

- Built DynaMimicGen, a DMP-based data generation framework that synthesizes large manipulation datasets from one or two human demonstrations instead of ten, improving generation success by up to 233% over MimicGen and lifting Diffusion Policy performance (e.g. Square: 75%→87%), validated in simulation and on a real Franka Emika Panda [P1].
- Developed DROM, a language-guided diffusion framework that learns 17 manipulation skills in a single policy and composes them via an LLM task decomposer into long-horizon instructions of up to 9 chained skills, raising average success from 61% to 91% over the Motion Planning Diffusion baseline on a FANUC CRX-25iA and a Franka Research 3 [P2].
- Designed personalized DMP trajectories with real-time velocity scaling for collaborative transport of an aircraft engine cowl lip, outperforming an industrial BiTRRT planner in user preference and in physiological stress measures (EEG and skin conductance) [P3].
- Benchmarked vision–language–action models (SmolVLA, OpenVLA) for manipulation, with State of the Art simulation frameworks, such as ROboSuite, ManiSkill.

Vision-based activity recognition for human–robot collaboration (*currently under review*)

- Collected a new multimodal RGB-D dataset of 19 assembly activities for the collaborative assembly of a heavy aircraft component and adapted a transformer-based few-shot action recognition method to this industrial task.
- Reached 95% recognition accuracy on 24 participants after training on 5 operators (30 trials per activity), showing strong cross-person generalization from limited data.
- Integrated recognition outputs into a Hierarchical Task Network (HTN) planner controlling a FANUC CRX-25iA cobot, enabling the robot to anticipate the operator's next assembly step in real time.

EU-funded projects & system integration

- Contributed to the Horizon Europe project Fluently (22 industrial and academic partners), leading research development, experimentation and system integration for the Dexterity use case, where a robot collaborates with a human during non-expert operator training.
- Integrated learning from demonstration, speed scaling based on human-robot distance, natural-language intent recognition and 6D object pose estimation (FoundationPose) into a complete collaborative robotic cell (CRX20ia-L) using ROS 2.
- Presented the system in a live human–robot collaboration demonstration at the project's final General Assembly [video]; supervised 1 MSc student.

SELECTED PUBLICATIONS

- **Pomponi, V.**, Franceschi, P., Baraldo, S., Avram, O., Roveda, L., Gambardella, L.M., Valente, A.: Dynamimicgen: A data generation framework for robot learning of dynamic tasks. *IEEE Robotics and Automation Letters* (2026).

- **Pomponi, V.**, Felici, R., Franceschi, P., Baraldo, S., Avram, O., Roveda, L., Gambardella, L.M. and Valente, A., 2026. DROM: A Language-Guided Diffusion Framework for Multi-Skill Robotic Manipulation. arXiv preprint arXiv:2609.37348.
- Franceschi, P., Bussolan, A., **Pomponi, V.**, Avram, O., Baraldo, S., & Valente, A. (2025). Human-robot collaborative transport personalization via Dynamic Movement Primitives and velocity scaling. 2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), 1235-1242.
- Falerni, M. M., **Pomponi, V.**, Karimi, H. R., Nicora, M. L., Dao, L. A., Malosio, M., & Roveda, L. (2024). A framework for human–robot collaboration enhanced by preference learning and ergonomics. Robotics and Computer-Integrated Manufacturing, 89, 102781.

TEACHING

Teaching Assistant — Diffusion Models

Spring 2026

Politecnico di Milano, Italy · Mechanical Engineering Application of Deep Learning · MSc level

- Prepared and delivered lectures, exercises sessions, and Python labs on diffusion models (topics, e.g. forward/reverse diffusion processes, ELBO, derivation of training objective, DDPM, score-based models, conditioning, applications to robot learning of state of the art Diffusion Policy) for 100 students.
- Created and maintain an open-source educational repository on diffusion models, with theory notes and runnable code examples: [GitHub](#) (3 ★).

Teaching Assistant — Robotics

Autumn 2026

SUPSI, Lugano, Switzerland · Robotics and Perception · BSc level

- Taught lectures and lab sessions on ROS 2, robot kinematics and computer vision to 25 students, supervising hands-on robotics projects and grading assignments

Teaching Assistant — Physics

2025 – 2026, 2026 – 2027

SUPSI, Lugano, Switzerland · Physics · BSc level

- Taught lectures and lab sessions on first-year physics, covering classical mechanics (kinematics, Newton's laws, work, energy and momentum, rotational dynamics), fluids, and thermodynamics.
- Prepared problem sets and mini-quizzes, led written-exam preparation sessions, and graded assignments and exams.

EDUCATION

PhD in Informatics

2023 – Expected September 2027

Università della Svizzera italiana (USI), Lugano, Switzerland

- Thesis: diffusion-based generative models for robotic manipulation and human–robot interaction.
- Advisors: Luca Maria Gambardella, Anna Valente, and Stefano Baraldo.

MSc in Mechanical Engineering

2016 – 2022

Politecnico di Milano, Italy · Final grade 106/110

- Thesis: Implementation of Active multi-Preference Learning for a collaborative robot with ergonomics assessment [P4].
- Advisors: Hamid-Reza Karimi and Loris Roveda

TECHNICAL SKILLS

- **Machine Learning & GenAI:** Python, PyTorch, diffusion models (e.g. Diffusion Policy, Implicit Reactive Diffusion Policy), imitation learning, vision–language–action models (e.g. OpenVLA, smolVLA), transformers, preference learning
- **Robotics:** ROS/ROS2, Dynamic Movement Primitives, manipulation, human–robot collaboration, robots: e.g. Franka, FANUC CRX20ia-L, FANUC CRX25ia, simulators: e.g. MuJoCo, Isaac Sim, simulation frameworks: RoboSuite, RoboMimic, ManiSkill.
- **Computer Vision:** Activity recognition, human pose estimation, RGB-D perception, OpenCV
- **Tools:** Git, Linux, Docker, Weights & Biases, CUDA, GPU cluster training

LANGUAGES

ADDITIONAL

- Reviewer for ICRA / IROS / RA-L / IJRR